WHERE THINGS STAND AFTER THE SIXTH ROUND

Labels: *proved* (complete argument from standard facts), *computed* (exact or high-precision computation with saved code, independently rechecked where stated), *plausible*, *conjecture*. Records are in `notes/R6/`; the companion note is `drop.pdf`.

1. Notation

$Y_r = S^3_r(K)$. On $Y_{\pm 2}$ there are two SO(3)-bundles, P_0 ($w_2 = 0$) and P_1 ($w_2 \neq 0$). For an irreducible flat connection α, its level is $c(\alpha) = \widetilde{\mathrm{CS}} - \widetilde{\mathrm{gr}}/8 = (3 + \rho(\alpha))/16$, with ρ the adjoint ρ-invariant; $\lambda_i(Y)$ is the least level of a nonzero class of $I(Y, P_i)$, so $\lambda_0 = \ell$ (DLME). The *both-bundle drop* is $\min(\lambda_0, \lambda_1)(Y_2) - \min(\lambda_0, \lambda_1)(Y_{-2})$; the *mixed drops* are $m_1 = \lambda_0(Y_2) - \lambda_1(Y_{-2})$ and $m_2 = \lambda_1(Y_2) - \lambda_0(Y_{-2})$. A cosmetic pair makes all of these zero.

2. What the Floer-theoretic layer can and cannot do

No-go theorem (proved; five independent explicit models, all checked by code, with negative controls). The following inputs, all together, cannot prove any of the drops above to be positive: the bundle-exchanging exact triangles (outer terms summed over both bundles, middle term the theory of S^3 plus the traceless theory of (S^3, K)) with the level of every map and homotopy; strict positivity of energy on every singular map; arbitrary differentials; the gradings mod 8, the twist χ, the null-homotopies, Euler characteristics and the Casson–Lin count, balanced gradings; the mirror triangles, the automorphisms forced by a cosmetic pair, and the lift to the double cover branched along S. What these inputs do not see is *where the generators sit* (their ρ-values) and *which generators survive* in homology.

What the layer does give (proved). (i) A strict drop from positivity alone, except in one precisely described configuration: the lowest classes of Y_2 (both bundles) are carried into the traceless theory, and every traceless class at or below that level comes from Y_2 and dies in Y_{-2}. A cosmetic pair forces this configuration at the bottom of the spectrum and its mirror image at the top. (ii) A counting form of DLME's argument: $\min(\lambda_0, \lambda_1)(Y_{-2}) \leq T_1 - \frac{1}{8}$, where T_1 is the first level at which the classes of $I(Y_2)$ (both bundles) born so far outnumber the traceless bars alive. (iii) The constant $\frac{1}{16}$ enters only as the ρ-term of the flat connection of $L(4,1)$ with holonomy a rotation by $\pi/2$; it appears only in maps that move classes *up*, so it is a ceiling on the cost of absorption into the traceless theory, never a floor. Every absorption cost in $(0, \frac{1}{16}]$ occurs in the models, so "absorption costs more than $\frac{1}{16}$" is equivalent to the conclusion and is not admissible.

Corrections to earlier notes (proved). The level c is not a filtration: the differential raises it by $\frac{1}{8} - E$; boundary depth must be measured accordingly. The vertex lemma of the triangle note was too weak by $\frac{1}{8}$. The hypothesis "no traceless representation sends the longitude to ± 1" is genuinely extra, even when $\Delta_K = 1$ (see §4).

3. The pillowcase reduction, and why its key lemma is withdrawn

The generators of $I(Y_{\pm 2})$ and of the traceless theory are the points where one curve $\mathcal{C}$ (the irreducible SU(2) representations of the knot group, in the pillowcase with meridian angle α and longitude angle β) meets three lines. Their levels are $c(Y_N, x) = \Phi(x) + \frac{3}{16}\operatorname{sgn} N - N\alpha(\frac{1}{2} - \alpha) - \frac{1}{8}\iota_N(x)$ with $d\Phi = 2(\alpha - \frac{1}{4})\, d\beta$ along $\mathcal{C}$ (computed exactly for torus knots, to 10^{-7} for the figure-eight knot; plausible in general). At $\alpha = \frac{1}{4}$ the slope term is $\pm(\frac{3}{16} - 2 \cdot \frac{1}{16}) = \pm\frac{1}{16}$, which is where $\frac{1}{16}$ comes from on the level of generators.

Correction. The rule for ι_N printed in `drop.pdf` has the wrong sign for $N < 0$; the consistent rule is $\iota_N = \operatorname{sgn} N$ when the slope of $\mathcal{C}$ lies strictly between 0 and $-N$, and 0 otherwise (computed: forced by $c(-Y) = \frac{3}{8} - c(Y)$ and by the exact mirror-torus data; gives Euler characteristic 0 where required).

Withdrawn. Lemma 5.1 of `drop.pdf` derived a drop in the least generator level from a *steepness* hypothesis on $\mathcal{C}$ near the minimum of Φ. Steepness bounds how much Φ, i.e. the Chern–Simons value, can change between neighbouring generators: it is an energy hypothesis in disguise, and is withdrawn as a strategy. It also fails in practice: its hypotheses hold on only 3 of 48 components in the tests below.

4. Tests on knots with Alexander polynomial one (computed)

The curve $\mathcal{C}$ was computed numerically for the Kinoshita–Terasaka knot (genus 2), the Conway knot, the pretzel knot $P(-3, 5, 7)$ and the untwisted Whitehead doubles $D_\pm$ of the right-handed trefoil, in both chiralities; the pipeline reproduces the figure-eight and torus-knot values to 10^{-7}. Since the constants of integration of Φ on different closed components are unknown, only a range for the drop in the least generator level (both bundles) is determined:

knot	as given	mirror
Kinoshita–Terasaka, Conway (same curve)	[0.096, 0.119]	[0.111, 0.187]
D_+ (right-handed trefoil)	[0.085, 0.125]	[0.109, 0.125]
D_- (right-handed trefoil)	[0.100, 0.125]	[0.099, 0.125]
$P(-3,5,7)$	[0, 0.164]	[0, 0.123]

So the generator-level drop exceeds $\frac{1}{16}$ for Kinoshita–Terasaka, Conway and both Whitehead doubles. For $P(-3,5,7)$ the representation variety is degenerate: four flat loops lie on $\beta = 0$, carrying twisted generators at $(\alpha,\beta) = (\frac{1}{4},0)$ that are shared by Y_2 and Y_{-2} with level difference exactly 0; a drop of 0 is not excluded. On the Whitehead doubles all traceless points also sit at $(\frac{1}{4},0)$ (longitude $\mapsto 1$), as Morse–Bott circles; they are not binary dihedral. Thus traceless representations sending the longitude to ± 1 occur for knots with $\Delta_K = 1$, and they are exactly where the generator-level approach gives nothing. (For Kinoshita–Terasaka and Conway, Y_2 and Y_{-2} even have different numbers of irreducible flat connections, 48 against 32, which distinguishes them by counting; this says nothing in general.)

5. **Other data (computed)**

The $\pm 1/n$ surgeries on torus knots (two independent codes, 282 torus knots of both chiralities, $n \le 3$; 18 knots up to $n = 8$). $\ell(Y_{-1/n})$ decreases in n, as in DLME's chain. The one-step drops have no positive universal bound: $\ell(Y_{-1}) - \ell(Y_{-1/2})$ has minimum $7/232 \approx 0.030$ (mirror $T(3,5)$), below $\frac{1}{32}$. The two-step drop satisfies $\ell(Y_{-1/n}) - \ell(Y_{-1/(n+2)}) \ge \frac{1}{8}$ in every case, with equality exactly for mirror torus knots with p, q both odd, at odd n. The equality comes from a flat SO(3) connection on the composite cobordism (intersection form $-I_2$, a sphere of square -2 with boundary $\mathbb{R}P^3$) whose $\mathbb{R}P^3$-end is the rotation by π, at representations with meridian and longitude both traceless; the term $\frac{1}{8}$ is that end's ρ-contribution. The triangle argument cannot prove the two-step bound: its middle term absorbs every class.

Census. Among all 313,230 prime knots with at most 15 crossings and 3,116 census knots, the only knots whose ± 2 surgeries are both Seifert fibred are the torus knots and the figure-eight knot (verified by interval arithmetic through 14 crossings and for the census; at 15 crossings, 243,282 knots are excluded on floating-point slope lengths above 12).

6. **What is running and what remains**

Running. Fibres in Brieskorn spheres that are not torus knot surgeries (no $a_k \equiv \pm 1 \bmod a_i a_j$): for each singular fibre K of $\Sigma(a_1,a_2,a_3)$, compute ℓ of the $-1, -\frac{1}{2}, -\frac{1}{3}, -\frac{1}{4}$ surgeries (and Σ, and $+\frac{1}{n}$), and test whether these decrease and whether $\ell(Y_{-1}) - \ell(Y_{-1/3}) \ge \frac{1}{8}$; if not, find the bound.

Remaining, as we now see it. (i) An input outside the no-go list, not an energy estimate, governing which generators survive or where they sit. (ii) The traceless representations sending the longitude to ± 1, which $P(-3,5,7)$ shows can occur, degenerately, when $\Delta_K = 1$; any proof must handle them. (iii) The general formula for c on the pillowcase.